

Explore following 4 websites and answer the questions

- bitcoin.org
- Blockchain.com
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Bitcoin was introduced by Satoshi Nakamoto.
2. Bitcoin transactions are recorded on which data structure?
Bitcoin transactions are recorded on a distributed ledger known as a blockchain, which organizes data into a sequential chain of blocks.
3. Bitcoin uses which consensus mechanism?
Bitcoin secures its network and validates transactions using the Proof of Work consensus mechanism.
4. The smallest unit of Bitcoin is called?
The smallest unit of a Bitcoin is called a satoshi (equal to 0.00000001 BTC).
5. Bitcoin is a **peer-to-peer** type of what system?
Bitcoin is a peer-to-peer electronic cash system.

Wallets

6. A wallet stores the user's **private** key.
7. The recovery phrase used in wallets is called a **seed** phrase.

8. Hardware wallets store keys **offline** (online/offline).
9. Bitcoin addresses are derived from **public** keys.
10. What is required to sign a transaction? (public/private)
private key

Security

11. Bitcoin prevents double-spending using **blockchain consensus** .
12. Bitcoin is secured using **asymmetric** cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin **Core**.
14. The process of verifying transactions is called **validation**.
15. The Bitcoin network is described as **decentralized**(centralized/decentralized).

Mining

16. Miners solve **cryptographic** puzzles to add blocks.
17. Mining rewards a block with newly created **bitcoins**.
18. The block time target for Bitcoin is **10** minutes.
19. Mining difficulty adjusts every **2016** blocks.
20. Miners compete to find a valid block **hash** (hash/signature).

Blockchain Structure

21. Each block contains a **Merkle** root.
22. Blocks are linked using the previous block's **hash**.
23. A Bitcoin transaction contains inputs and **outputs**.
24. Unspent transaction outputs are called **Unspent Transaction Outputs**.

25. The first-ever Bitcoin block is called the **Genesis** block.

Network

26. A full node stores the entire blockchain **ledger**.

27. Bitcoin uses which protocol for communication? **P2P** (TCP/P2P)

28. A light wallet uses **SPV** mode (SPV/full).

29. Nodes relay information across a **P2P** network.

30. The maximum Bitcoin supply is **21** million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 961415
- Block hash:
000000000000000000000000deb8e23160cb5f8d577873475dc5d3e76e466f086a29
- Timestamp: 2026-08-07 15:40:12
- Number of transactions: 4039
- Miner/pool name: AntPool

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction

d7b9df8d9239464ef54e30454b9c363730509d061207f3d2a3002dd15bc7bc29

11. Number of inputs: 1

12. Number of outputs: 7

13. Transaction fee: 0.00000000 BTC (\$0.00)

14. How many confirmations does it have? 36

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

39C7fxSzEACPjM78Z7xdPxhf7mKxJwvfMJ

16. Record:

- Total BTC received: 85,042.75308098 BTC
- Total BTC sent: 84,801.39441522 BTC
- Current balance: 187.94541811 BTC

What is the role of a full node?

A **full node** stores the complete Bitcoin blockchain, independently verifies every transaction and block according to Bitcoin's consensus rules, relays valid transactions to other nodes, and helps keep the Bitcoin network secure and decentralized.

List any two developer resources available on bitcoin.org

1. **Bitcoin Core**
2. **Running a Full Node** (Developer Guide)

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: \$64,938.44
- 24-hour % change: 1.64%
- Market capitalization: \$1.3 Trillion
- Circulating supply: 20.06 Million BTC
- Total/maximum supply: 21 Million BTC

Task C2: Historical Data

- What was the price of Bitcoin one year ago?
\$64,933.03
- Note the highest and lowest prices in the last 30 days.

Lowest: \$61,630.55 Highest: \$66,855.57

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price changes rapidly due to factors such as market demand and supply, investor sentiment, news, government regulations, macroeconomic events, and speculation. Since it is not controlled by a central authority, its value can fluctuate significantly within a short period.

List two investment risks mentioned.

High price volatility, which can lead to significant financial losses.

Regulatory uncertainty, as changes in government laws and regulations can affect Bitcoin's price and adoption.

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave me the clearest understanding because it explains Bitcoin, blockchain, mining, wallets, and full nodes in a simple, beginner-friendly manner with accurate technical information. It is easy to navigate and focuses on the fundamentals of how Bitcoin works.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transaction records on a decentralized network of computers called nodes. Every transaction is verified using cryptographic techniques and validated through a consensus mechanism before being added to the blockchain. Each block contains the hash of the previous block, making the chain secure and resistant to tampering. Since every participant has a copy of the ledger and no single entity controls the network, it is extremely difficult to alter or manipulate transaction data, ensuring transparency, security, and trust among users.